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Checking the Cox Proportional Hazards Model with Interval-Censored Data

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ABSTRACT

This article presents a general framework for checking the adequacy of the Cox proportional hazards model with interval-censored data, which arise when the event of interest is known only to occur over a random time interval. Specifically, we construct certain stochastic processes that are informative about various aspects of the model, that is, the functional forms of covariates, the exponential link function and the proportional hazards assumption. We establish their weak convergence to zero-mean Gaussian processes under the assumed model through empirical process theory. We then approximate the limiting distributions by Monte Carlo simulation and develop graphical and numerical procedures to check model assumptions and improve goodness of fit. We evaluate the performance of the proposed methods through extensive simulation studies and provide an application to the Atherosclerosis Risk in Communities Study. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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Goodness of fit; Interval censoring; Model misspecification; Profile likelihood; Transformation

1. Introduction

The Cox (1972) proportional hazards model is commonly used to analyze potentially censored event time data arising frequently in clinical, epidemiological, financial and sociological studies. This model specifies that the hazard function for the event time T conditional on a p-vector of possibly time-dependent covariates $\boldsymbol{X}(\cdot)$ takes the form

$$\lambda(t \mid \boldsymbol{X}) = \lambda(t) \exp\{\boldsymbol{\beta}^{\mathrm{T}} \boldsymbol{X}(t)\}, \qquad (1)$$

where $\boldsymbol{\beta}$ is a vector of unknown regression parameters, and $\lambda(\cdot)$ is an arbitrary baseline hazard function. The estimation for the regression parameters $\boldsymbol{\beta}$ and the cumulative baseline hazard function $\Lambda(t) = \int_0^t \lambda(u)\mathrm{d}u$ has been studied for both right-censored data (Breslow 1972; Cox 1975; Andersen and Gill 1982) and interval-censored data (Huang 1996; Huang and Wellner 1997; Zeng, Mao, and Lin 2016).

Although model (1) possesses some flexibility from its semiparametric form, it can fail in three ways (Lin, Wei, and Ying 1993). First, the proportional hazards assumption does not hold, that is, the hazard ratios are time-varying. Second, the functional forms or transformations of individual covariates in the exponent of the model are not correctly specified. Third, the exponential form of the multiplier or link function is inappropriate. The model misspecification can result in invalid statistical inference and misleading analysis results. For example, violating the proportional hazards assumption can result in a significant loss of statistical efficiency (Lagakos and Schoenfeld 1984) and incorrect variance estimators (DiRienzo and Lagakos 2002). Likewise, misspecifying the transformation of a continuous covariate would yield incorrect characterization of the covariate effect and loss of statistical efficiency (Lagakos 1988). Omitting relevant covariates or using an inappropriate link function would yield biased estimators for the regression parameters (Gail, Wieand, and Piantadosi 1984; Struthers and Kalbfleisch 1986; Kosorok, Lee, and Fine 2004). Violations of the assumptions have been widely observed in both clinical trials (van Laar et al. 2014; Holme et al. 2014) and cohort studies (Lin et al. 2022a, 2022b).

Graphical and numerical methods for assessing the adequacy of model (1) with right-censored data have been extensively studied; see Lin, Wei, and Ying (1993) and the references therein. In contrast, there do not exist any methods for checking model (1) with interval-censored data. Farrington (2000) proposed a series of residual plots to check parametric proportional hazards models with interval-censored data. No asymptotic distributions have been derived for such residual plots, making it difficult to determine whether an observed residual pattern reflects model misspecification or is a phenomenon likely to occur even if the model is correctly specified.

It is highly challenging to check the adequacy of model (1) with interval-censored data for several reasons. First, as the prerequisites for model checking, the estimation of model (1) with interval-censored data is already very challenging. This problem was not satisfactorily resolved until Zeng, Mao, and Lin (2016) devised an EM algorithm to compute the nonparametric maximum likelihood estimators $\widehat{\boldsymbol{\beta}}$ and $\widehat{\Lambda}$ for $\boldsymbol{\beta}$ and Λ, respectively. Second, the martingale residuals (Barlow and Prentice 1988; Therneau, Grambsch, and Fleming 1990) are not computable under interval censoring, such that the model-checking methods commonly used for right-censored data are not applicable to interval-censored data. Third, $\widehat{\Lambda}$ converges to Λ at a rate slower than $n^{-1/2}$ (Zeng, Mao, and Lin 2016), hindering the investigation of the asymptotic behavior for any test statistics that contain $\widehat{\Lambda}$.

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In this article, we propose a class of graphical and numerical procedures for checking model (1) with general interval-censored data. Specifically, we construct a class of stochastic processes that are informative about different aspects of the model, that is, the functional form of a covariate, the exponential link function and the proportional hazards assumption. We use empirical process theory (van der Vaart and Wellner 1996) to show that, under the null hypothesis of no model misspecification, the proposed stochastic processes converge weakly to zero-mean Gaussian processes whose distributions can be approximated through Monte Carlo simulation. These results enable one to compare each observed process with a number of realizations from its null distribution and assess how unusual the observed pattern is.

2. Methods

To motivate our approach, we consider the case of no censoring, that is, the event time T being always observed. Define the counting process $N(t) = I(T \le t)$, where $I(\cdot)$ is the indicator function. If model (1) is correctly specified, $M(t) = N(t) - \int_0^t I(T \ge u)\exp\{\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{X}(u)\}\mathrm{d}\Lambda(u)$ is a martingale. The stochastic processes proposed by Lin, Wei, and Ying (1993) for checking model (1) are the empirical averages of the following statistics: $\int_0^\infty I\{X_j(u) \le x\}\mathrm{d}M(u)$ for checking the functional form of the jth covariate X_j; $\int_0^\infty I\{\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{X}(u) \le x\}\mathrm{d}M(u)$ for checking the exponential link function; $\int_0^y X_j(u)\mathrm{d}M(u)$ for checking the proportional hazards assumption with respect to X_j; and $\int_0^y I\{\boldsymbol{X}(u) \le x\}\mathrm{d}M(u)$ for checking the overall fit of the model. All these statistics can be written in a general form

$$\int_0^\infty g\{\boldsymbol{X}(u); \boldsymbol{\beta}, x, y\}\mathrm{d}M(u), \tag{2}$$

where $g\{\boldsymbol{X}(u); \boldsymbol{\beta}, x, y\}$ pertains to $I\{X_j(u) \le x\}$, $I\{\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{X}(u) \le x\}$, $X_j(u)I(u \le y)$ and $I\{\boldsymbol{X}(u) \le x\}I(u \le y)$ for the aforementioned four processes. The test statistics in (2) are equivalent to the score statistics for testing the null hypothesis of $\gamma = 0$ under the expanded Cox model

$$\lambda(t \mid \boldsymbol{X}) = \lambda(t)\exp[\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{X}(t) + \gamma g\{\boldsymbol{X}(t); \boldsymbol{\beta}, x, y\}]. \tag{3}$$

Remark 1. The equivalence of the statistic in (2) and the score test statistic is proved in Appendix A. The statistic in (2) measures the correlation between the g-function and the martingale $M(\cdot)$, where the g-function is designed to check a specific aspect of the model, and $M(\cdot)$ pertains to the difference between the observed and model-predicted numbers of events over time. Thus, (2) is informative about the potential misspecification of a specific aspect of the model. As shown in Appendix B, proper choices of the g-function lead to consistent tests against the alternatives of interest. For this reason, the g-function is allowed to depend on $\boldsymbol{\beta}$ in order to accommodate $I\{\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{X}(u) \le x\}$, which is designed for checking the exponential link function.

2.1. General Test Statistics

The data consist of n independent observations $O_i = (L_i, R_i, \boldsymbol{X}_i)$ $(i = 1, \ldots, n)$, where $(L_i, R_i]$ is the smallest interval containing the event time T_i of the ith subject. The interval endpoints L_i and R_i arise from a random sequence of examination times, denoted by $0 = U_{i0} < U_{i1} < \cdots < U_{iK_i} < U_{i,K_i+1} = \infty$. We assume that the examination times are independent of T_i conditional on $\boldsymbol{X}_i(\cdot)$. Let $0 = t_0 < t_1 < \cdots < t_m$ denote the set consisting of 0 and the unique values of $L_i > 0$ and $R_i < \infty$ $(i = 1, \ldots, n)$. The nonparametric maximum likelihood estimator $\widehat{\Lambda}$ is a step function that jumps only at these time points, with respective jump sizes of $\lambda_0 = 0, \lambda_1, \ldots, \lambda_m$. We obtain $(\widehat{\boldsymbol{\beta}}, \widehat{\Lambda})$ using the EM algorithm of Zeng, Mao, and Lin (2016).

We fix $\boldsymbol{\beta}$ at $\widehat{\boldsymbol{\beta}}$ and consider the expanded likelihood function under model (3)

$$\begin{aligned} &L_n(\gamma, \Lambda; \widehat{\boldsymbol{\beta}}, x, y) \\ &= \prod_{i=1}^n \left\{ \exp\left(-\int_0^{L_i} \exp[\widehat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{X}_i(u) + \gamma g\{\boldsymbol{X}_i(u); \widehat{\boldsymbol{\beta}}, x, y\}]\mathrm{d}\Lambda(u)\right) \right. \\ &\qquad \left. - \exp\left(-\int_0^{R_i} \exp[\widehat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{X}_i(u) + \gamma g\{\boldsymbol{X}_i(u); \widehat{\boldsymbol{\beta}}, x, y\}]\mathrm{d}\Lambda(u)\right) \right\}, \end{aligned}$$

where the second integral is set to ∞ if $R_i = \infty$ to accommodate right-censored observations. To obtain the score statistics for testing the null hypothesis of $\gamma = 0$, we calculate the profile log-likelihood function

$$\mathrm{pl}_n(\gamma; \widehat{\boldsymbol{\beta}}, x, y) = \max_{\Lambda} \log L_n(\gamma, \Lambda; \widehat{\boldsymbol{\beta}}, x, y)$$

for any γ in a neighborhood of 0. Specifically, we maximize

$$\begin{aligned} \prod_{i=1}^n \Bigg(&\exp\Bigg[-\sum_{t_q \le L_i} \lambda_q \exp\{\widehat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{X}_{iq} + \gamma g(\boldsymbol{X}_{iq}; \widehat{\boldsymbol{\beta}}, x, y)\}\Bigg] \\ &- I(R_i < \infty)\exp\Bigg[-\sum_{t_q \le R_i} \lambda_q \exp\{\widehat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{X}_{iq} + \gamma g(\boldsymbol{X}_{iq}; \widehat{\boldsymbol{\beta}}, x, y)\}\Bigg]\Bigg) \end{aligned} \tag{4}$$

with $\widehat{\boldsymbol{\beta}}$ and γ fixed, where $\boldsymbol{X}_{iq} = \boldsymbol{X}_i(t_q)$ $(q = 1, \ldots, m)$. To make the maximization tractable, we introduce mutually independent Poisson random variables W_{iq} with means $\lambda_q \exp\{\widehat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{X}_{iq} + \gamma g(\boldsymbol{X}_{iq}; \widehat{\boldsymbol{\beta}}, x, y)\}$. In addition, let $A_i = \sum_{t_q \le L_i} W_{iq}$ and $B_i = I(R_i < \infty)\sum_{L_i < t_q \le R_i} W_{iq}$. Because the joint probability of $A_i = 0$ and $B_i > 0$ is $\exp[-\sum_{t_q \le L_i} \lambda_q \exp\{\widehat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{X}_{iq} + \gamma g(\boldsymbol{X}_{iq}; \widehat{\boldsymbol{\beta}}, x, y)\}] - I(R_i < \infty)\exp[-\sum_{t_q \le R_i} \lambda_q \exp\{\widehat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{X}_{iq} + \gamma g(\boldsymbol{X}_{iq}; \widehat{\boldsymbol{\beta}}, x, y)\}]$, the likelihood for the observed data $\{A_i = 0, B_i > 0 : i = 1, \ldots, n\}$ is the same as (4). Thus, we implement an EM-type algorithm by treating the W_{iq} as latent variables, wherein the M-step is tantamount to maximizing a weighted sum of Poisson log-likelihood functions and has closed-form solutions for λ_q $(q = 1, \ldots, m)$.

In the E-step, we calculate the conditional mean of W_{iq} given the observed data:

$$\begin{aligned} &\widehat{E}(W_{iq}) \\ &= \frac{I(L_i < t_q \le R_i < \infty)\lambda_q \exp\{\widehat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{X}_{iq} + \gamma g(\boldsymbol{X}_{iq}; \widehat{\boldsymbol{\beta}}, x, y)\}}{1 - \exp[-\sum_{L_i < t_{q'} \le R_i} \lambda_{q'} \exp\{\widehat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{X}_{iq'} + \gamma g(\boldsymbol{X}_{iq'}; \widehat{\boldsymbol{\beta}}, x, y)\}]}. \end{aligned}$$

In the M-step, we update

$$\lambda_q = \frac{\sum_{i=1}^{n} I(R_i^* \geq t_q)\widehat{E}(W_{iq})}{\sum_{i=1}^{n} I(R_i^* \geq t_q)\exp\{\widehat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{X}_{iq} + \gamma g(\boldsymbol{X}_{iq}; \widehat{\boldsymbol{\beta}}, x, y)\}}$$

for $q = 1, \ldots, m$, where $R_i^* = I(R_i < \infty)R_i + I(R_i = \infty)L_i$. Starting with $\widehat{\Lambda}$ as the initial value, we iterate between the E-step and the M-step until the maximum absolute difference between the parameter estimates of two successive iterations is less than a certain threshold to obtain the profile log-likelihood maximizer $\widetilde{\Lambda}(\cdot; \widehat{\boldsymbol{\beta}}, \gamma, x, y)$. By the definition of the profile log-likelihood function, $\mathrm{pl}_n(\gamma; \widehat{\boldsymbol{\beta}}, x, y) = \log L_n\{\gamma, \widetilde{\Lambda}(\cdot; \widehat{\boldsymbol{\beta}}, \gamma, x, y); \widehat{\boldsymbol{\beta}}, x, y\}$.

The proposed test statistic is defined as the $n^{-1/2}$-scaled first-order numerical derivative of the profile log-likelihood function with respect to γ at $\gamma = 0$, given by

$$W_n(x, y) = n^{-1/2}\frac{\mathrm{pl}_n(\delta_n; \widehat{\boldsymbol{\beta}}, x, y) - \mathrm{pl}_n(-\delta_n; \widehat{\boldsymbol{\beta}}, x, y)}{2\delta_n},$$

where δ_n is the step size in the order $n^{-1/2}$ for the numerical difference. The proposed test statistic, $W_n(x, y)$, represents the score statistics for testing the null hypothesis of $\gamma = 0$ under model (3). By the equivalence between the score statistics of the Cox model and the martingale transformations in (2), $W_n(x, y)$ is informative about model misspecification and thus can be used to check the adequacy of model (1). In addition, $W_n(x, y)$ has several attractive features. First, $W_n(x, y)$ is free of unknown parameters by fixing $\boldsymbol{\beta}$ at $\widehat{\boldsymbol{\beta}}$ and thus computable. Second, $W_n(x, y)$ is expected to be asymptotically normal because $n^{-1/2}W_n(x, y)$ is an empirical approximation to the efficient score function for γ in model (3) with known $\boldsymbol{\beta}$. Because the efficient score function for γ is orthogonal to the tangent space spanned by the score functions for Λ, the estimation of Λ will impact $W_n(x, y)$ only in the second order. In the current context, the convergence rate of $\widehat{\Lambda}$ is slower than $n^{-1/2}$ but faster than $n^{-1/4}$ (Zeng, Mao, and Lin 2016), implying that the impact of $\widehat{\Lambda}$ on $W_n(x, y)$ is of an order smaller than $n^{-1/2}$. Therefore, the asymptotic distribution of $W_n(x, y)$ depends only on the $n^{1/2}$-consistent estimator for the finite-dimensional parameter, which is asymptotically normal. Third, $W_n(x, y)$ accounts for the uncertainty associated with estimating $\boldsymbol{\beta}$ from the observed data and accommodates a broad class of estimators for $\boldsymbol{\beta}$. As implied by the proofs of Theorems 1 and 2, any regular and asymptotically linear estimator for $\boldsymbol{\beta}$ would result in a test statistic with the desired asymptotic properties.

Remark 2. The proposed test statistics are related to the Lagakos residuals used by Farrington (2000), which can be derived as the score functions of the regression parameters evaluated at the parameter estimators. Our test statistics are based on the efficient score functions of the regression parameters, where the infinite-dimensional parameter Λ is profiled out. This removes the effects of the slower convergence rate of $\widehat{\Lambda}$ and guarantees the asymptotic normality of the test statistics.

2.2. Asymptotic Theory

We establish the asymptotic properties of $W_n(x, y)$ under the following regularity conditions, wherein we omit the subscript i when referring to a random variable for a subject.

Condition 1. The true value of the regression parameters, $\boldsymbol{\beta}_0$, lies in the interior of a known compact set Θ in $\mathbb{R}^p$, and the true value of the cumulative baseline hazard function, $\Lambda_0(\cdot)$, is continuously differentiable with positive derivatives in $[0, \tau]$, the union of the supports of U_l $(l = 1, \ldots, K)$.

Condition 2. With probability one, the vector $\boldsymbol{X}(\cdot)$ has bounded total variation over $[0, \tau]$. In addition, if there exist a deterministic function $a_1(t)$ and a constant vector a_2 such that $a_1(t) + a_2^{\mathrm{T}}\boldsymbol{X}(t) = 0$ with probability 1, then $a_1(t) = 0$ for $t \in [0, \tau]$ and $a_2 = 0$.

Condition 3. The number of examination times K is positive with $E(K) < \infty$. The conditional probability $\Pr(U_K = \tau \mid K, \boldsymbol{X})$ is greater than some positive constant. In addition, $\Pr\{\min_{0 \leq l \leq K}(U_{l+1} - U_l) \geq \eta \mid K, \boldsymbol{X}\} = 1$ for some positive constant η. Finally, the densities of (U_l, U_{l+1}) conditional on $(K, \boldsymbol{X})$, denoted by $g_l(v, w \mid K, \boldsymbol{X})$ $(l = 0, \ldots, K)$, have continuous second-order partial derivatives with respect to u and v when $v - u \geq \eta$ and $u, v \in [0, \tau]$, and they are continuously differentiable functionals with respect to $\boldsymbol{X}$.

Condition 4. The domain of $W_n(x, y)$ is a bounded compact subset of $\mathbb{R}^2$, denoted by $\mathbb{T}$. With probability one, $g\{\boldsymbol{X}(t); \boldsymbol{\beta}, x, y\}$ is uniformly bounded with uniformly bounded total variation in $(t, \boldsymbol{\beta}, x, y) \in [0, \tau] \times B_{\epsilon_1}(\boldsymbol{\beta}_0) \times \mathbb{T}$ for some positive constant ϵ_1, where $B_\epsilon(\boldsymbol{\beta}_0) = \{\boldsymbol{\beta} : \|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| \leq \epsilon\}$. In addition, the joint distribution function of $\boldsymbol{X}(t)$ and $\int_0^t \exp\{\boldsymbol{\beta}_0^{\mathrm{T}}\boldsymbol{X}(s)\}g\{\boldsymbol{X}(s); \boldsymbol{\beta}, x, y\}\mathrm{d}\Lambda_0(s)$ is continuous with respect to $\boldsymbol{\beta}$ at $\boldsymbol{\beta}_0$ uniformly for $(t, x, y) \in [0, \tau] \times \mathbb{T}$, and it is continuously differentiable with respect to t, x, and y when $\boldsymbol{\beta}$ is fixed at $\boldsymbol{\beta}_0$.

Conditions 1–3 are standard assumptions for regression analysis of interval-censored data and entail the asymptotic properties of $\widehat{\boldsymbol{\beta}}$ and $\widehat{\Lambda}$ established in Zeng, Mao, and Lin (2016). Condition 4 allows the g-function to have discontinuous trajectories in $(t, \boldsymbol{\beta}, x, y)$, but the expectation of its integral must be continuous with respect to $\boldsymbol{\beta}$ and differentiable with respect to t, x, and y. This condition ensures the convergence of some integral transforms of the g-function and the smoothness of the least favorable direction (Murphy and van der Vaart 2000).

Let $\boldsymbol{l}_{\boldsymbol{\beta}_0}(\cdot)$ be the efficient score function for $\boldsymbol{\beta}$ in model (1), and $l_{\gamma_0}(\cdot; x, y)$ be the efficient score function for γ in model (3) with $\boldsymbol{\beta}$ fixed at $\boldsymbol{\beta}_0$. Write $\boldsymbol{l}_0(\cdot; x, y) = \{\boldsymbol{l}_{\boldsymbol{\beta}_0}^{\mathrm{T}}(\cdot), l_{\gamma_0}(\cdot; x, y)\}^{\mathrm{T}}$. Let $\boldsymbol{V}_0(x, y)$ be the covariance matrix of $\boldsymbol{l}_0(\cdot; x, y)$, which is partitioned as

$$\boldsymbol{V}_0(x, y) = \begin{bmatrix} \boldsymbol{V}_{\boldsymbol{\beta}_0\boldsymbol{\beta}_0} & \boldsymbol{V}_{\boldsymbol{\beta}_0\gamma_0}(x, y) \\ \boldsymbol{V}_{\gamma_0\boldsymbol{\beta}_0}(x, y) & V_{\gamma_0\gamma_0}(x, y) \end{bmatrix}$$

to conform with the dimensions of $\boldsymbol{\beta}$ and γ. We describe below the asymptotic quadratic expansion of the profile log-likelihood.

Theorem 1. Under Conditions 1–4 and the null hypothesis that model (1) is correctly specified, for any random sequence $\widetilde{\gamma}$ that converges to $\gamma_0 = 0$ in probability and any random sequence $\widetilde{\boldsymbol{\beta}}$ that satisfies $\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\| = O_{\mathrm{p}}(n^{-1/2})$,

$$\begin{aligned}\mathrm{pl}_n(\widetilde{\gamma};\widetilde{\boldsymbol\beta},x,y) &= \mathrm{pl}_n(0;\widetilde{\boldsymbol\beta},x,y) + \widetilde{\gamma}\sum_{i=1}^{n} l_{\gamma_0}(O_i;x,y)\\ &\quad - n\widetilde{\gamma}(\widetilde{\boldsymbol\beta}-\boldsymbol\beta_0)^{\mathrm T}\boldsymbol V_{\boldsymbol\beta_0\gamma_0}(x,y)\\ &\quad - \frac{1}{2}n\widetilde{\gamma}^2 V_{\gamma_0\gamma_0}(x,y) + o_{\mathrm p}(n\widetilde{\gamma}^2 + n^{1/2}\widetilde{\gamma}),\end{aligned}$$

where the convergence rate of the remainder term is uniform for $(x,y)\in\mathbb{T}$.

The proof of Theorem 1 makes use of the Taylor series expansion of the profile log-likelihood function, where careful arguments are needed because the g-function is not assumed to be differentiable with respect to $\boldsymbol\beta$. The derivative of the profile log-likelihood maximizer with respect to $(\boldsymbol\beta,\gamma)$ will converge to the least favorable direction, which is shown to be a smooth function. These results guarantee the Donsker properties of certain function classes and thus the uniform convergence rate of the remainder term in Theorem 1.

The asymptotic expansion for $\widehat{\boldsymbol\beta}$ is

$$n^{1/2}(\widehat{\boldsymbol\beta}-\boldsymbol\beta_0) = n^{-1/2}\sum_{i=1}^{n}\boldsymbol V^{-1}_{\boldsymbol\beta_0\boldsymbol\beta_0}\boldsymbol l_{\boldsymbol\beta_0}(O_i) + o_{\mathrm p}(1). \quad (5)$$

By incorporating (5) into Theorem 1, we obtain the asymptotic expansion for $W_n(x,y)$ in Theorem 2.

Theorem 2. Under Conditions 1–4 and the null hypothesis that model (1) is correctly specified,

$$W_n(x,y) = n^{-1/2}\sum_{i=1}^{n}\phi_0(O_i;x,y) + o_{\mathrm p}(1),$$

where

$$\phi_0(O_i;x,y) = l_{\gamma_0}(O_i;x,y) - \boldsymbol V^{\mathrm T}_{\boldsymbol\beta_0\gamma_0}(x,y)\boldsymbol V^{-1}_{\boldsymbol\beta_0\boldsymbol\beta_0}\boldsymbol l_{\boldsymbol\beta_0}(O_i),$$

and the convergence rate of the remainder term is uniform for $(x,y)\in\mathbb{T}$. In addition, $W_n(x,y)$ converges weakly to a tight zero-mean Gaussian process $W_0(x,y)$ with a covariance function of

$$\sigma\{(x_1,y_1),(x_2,y_2)\} = E\{\phi_0(O;x_1,y_1)\phi_0(O;x_2,y_2)\}.$$

Theorem 2 implies that $W_n(x,y)$ is essentially a sum of independent random variables. This expansion will reduce to the expansion derived by Lin, Wei, and Ying (1993) for right-censored data if $L_i = R_i$ or $R_i=\infty$. We show that $\phi_0(O_i;x,y)$ belongs to some Donsker class indexed by (x,y), which results in the weak convergence of $W_n(x,y)$ to $W_0(x,y)$ under the null hypothesis of no model misspecification.

The null distribution of $W_n(x,y)$ can be approximated by the zero-mean Gaussian process

$$\widehat{W}_n(x,y) = n^{-1/2}\sum_{i=1}^{n}\widehat{\phi}(O_i;x,y)G_i,$$

where G_i $(i=1,\ldots,n)$ are independent standard normal random variables, and

$$\widehat{\phi}(O_i;x,y) = \widehat{l}_{\gamma_0}(O_i;x,y) - \widehat{\boldsymbol V}^{\mathrm T}_{\boldsymbol\beta_0\gamma_0}(x,y)\widehat{\boldsymbol V}^{-1}_{\boldsymbol\beta_0\boldsymbol\beta_0}\widehat{\boldsymbol l}_{\boldsymbol\beta_0}(O_i).$$

Here, $\widehat{l}_{\gamma_0}(O_i;x,y)$, $\widehat{\boldsymbol l}_{\boldsymbol\beta_0}(O_i)$, $\widehat{\boldsymbol V}_{\boldsymbol\beta_0\gamma_0}(x,y)$ and $\widehat{\boldsymbol V}_{\boldsymbol\beta_0\boldsymbol\beta_0}$ are the empirical versions of $l_{\gamma_0}(O_i;x,y)$, $\boldsymbol l_{\boldsymbol\beta_0}(O_i)$, $\boldsymbol V_{\boldsymbol\beta_0\gamma_0}(x,y)$ and $\boldsymbol V_{\boldsymbol\beta_0\boldsymbol\beta_0}$, respectively. To obtain these estimators, we let $\mathrm{pl}_{ni}(\gamma;\boldsymbol\beta,x,y)$ be the contribution of the ith subject to $\mathrm{pl}_n(\gamma;\boldsymbol\beta,x,y)$ and estimate $\boldsymbol l_0(O_i;x,y)$ and $\boldsymbol V_0(x,y)$ by

$$\widehat{\boldsymbol l}_0(O_i;x,y) = D_{h_n}\mathrm{pl}_{ni}(0;\widehat{\boldsymbol\beta},x,y) \quad \text{and}$$

$$\widehat{\boldsymbol V}_0(x,y) = n^{-1}\sum_{i=1}^{n} D_{h_n}\mathrm{pl}_{ni}(0;\widehat{\boldsymbol\beta},x,y)^{\otimes 2},$$

respectively, where $\boldsymbol a^{\otimes 2} = \boldsymbol a\boldsymbol a^{\mathrm T}$. Here, D_{h_n} is the first-order numerical derivative for $(\boldsymbol\beta,\gamma)\in\mathbb{R}^{p+1}$ with perturbation constant h_n of order $n^{-1/2}$. For $\boldsymbol\theta\in\mathbb{R}^{p+1}$ and perturbation constant h,

$$D_h f(\boldsymbol\theta) = \left\{\frac{f(\boldsymbol\theta + h\boldsymbol e_i) - f(\boldsymbol\theta - h\boldsymbol e_i)}{2h}\right\}_{i=1,\ldots,p+1},$$

where $\boldsymbol e_i$ is the ith canonical vector in $\mathbb{R}^{p+1}$. Consequently, $\widehat{l}_{\gamma_0}(O_i;x,y)$, $\widehat{\boldsymbol l}_{\boldsymbol\beta_0}(O_i)$, $\widehat{\boldsymbol V}_{\boldsymbol\beta_0\gamma_0}(x,y)$, and $\widehat{\boldsymbol V}_{\boldsymbol\beta_0\boldsymbol\beta_0}$ are the corresponding subvectors or submatrices of $\widehat{\boldsymbol l}_0(O_i;x,y)$ and $\widehat{\boldsymbol V}_0(x,y)$.

To approximate the null distribution of $W_n(x,y)$, we obtain a number of realizations from $\widehat{W}_n(x,y)$ by repeatedly generating normal random samples $\{G_i : i=1,\ldots,n\}$ with the empirical estimators for $\boldsymbol l_0(O_i;x,y)$ and $\boldsymbol V_0(x,y)$ fixed. Theorem 3 states that the conditional distribution of $\widehat{W}_n(x,y)$ given the observed data is the same in the limit as the unconditional distribution of $W_n(x,y)$ and thus justifies the use of $\widehat{W}_n(x,y)$.

Theorem 3. Under Conditions 1–4 and the null hypothesis that model (1) is correctly specified, $\widehat{W}_n(x,y)\mid O_1,\ldots,O_n$ converges weakly to $W_0(x,y)$.

The proofs of Theorem 1–3 are relegated to the supplementary material.

Remark 3. The standard normal random samples $\{G_i : i = 1,\ldots,n\}$ can be replaced by any independent and identically distributed random variables with mean 0 and variance 1 because the Donsker properties of $\widehat{\phi}(O;x,y)$ guarantee that the resulting $\widehat{W}_n(x,y)$, conditional on the observed data, continues to converge weakly to $W_0(x,y)$. However, since $W_0(x,y)$ is a zero-mean Gaussian process, standard normal random samples are the most natural choice over other distributions.

2.3. Model-Checking Procedures

We propose to check various aspects of model (1) by considering some special cases of the g-function. To check the functional form of the jth covariate, we let $g\{\boldsymbol X_i(t);\widehat{\boldsymbol\beta},x,y\} = I\{X_{ji}(t)\le x\}$, specifying model (3) as

$$\lambda(t\mid\boldsymbol X) = \lambda(t)\exp[\boldsymbol\beta^{\mathrm T}\boldsymbol X(t) + \gamma I\{X_j(t)\le x\}].$$

Under this specification, the general stochastic processes $W_n(x,y)$ and $\widehat{W}_n(x,y)$, which are derived based on the score statistics for testing $\gamma=0$, reduce to stochastic processes indexed only by x, denoted as $W_{jn}(x)$ and $\widehat{W}_{jn}(x)$, respectively. According to the general results of Section 2.2, the null distribution of $W_{jn}(\cdot)$ can be approximated by simulating the corresponding zero-mean Gaussian process $\widehat{W}_{jn}(\cdot)$. To assess how unusual the observed pattern of $W_{jn}(\cdot)$ is under model (1), we can plot a few, say 20, realizations from $\widehat{W}_{jn}(\cdot)$ along with the observed $W_{jn}(\cdot)$.

We complement the graphical inspection with the supremum test $\sup_x |W_{jn}(x)|$. Its p-value is obtained by generating a large number of realizations from $\sup_x |\widehat{W}_{jn}(x)|$ and comparing them to the observed value of $\sup_x |W_{jn}(x)|$.

To check the exponential link function, we let $g\{\boldsymbol{X}_i(t); \widehat{\boldsymbol{\beta}}, x, y\} = I\{\widehat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{X}_i(t) \leq x\}$, specifying model (3) as

$$\lambda(t \mid \boldsymbol{X}) = \lambda(t)\exp[\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{X}(t) + \gamma I\{\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{X}(t) \leq x\}].$$

We denote the resulting $W_n(x, y)$ and $\widehat{W}_n(x, y)$ as $W_{n,l}(x)$ and $\widehat{W}_{n,l}(x)$, respectively. Graphical and numerical inspections can be conducted by comparing $W_{n,l}(x)$ and $\widehat{W}_{n,l}(x)$ in the same way as comparing $W_{jn}(x)$ and $\widehat{W}_{jn}(x)$.

To check the proportional hazards assumption with respect to the jth covariate, we let $g\{\boldsymbol{X}_i(t); \widehat{\boldsymbol{\beta}}, x, y\} = X_{ji}(t)I(t \leq y)$, specifying model (3) as

$$\lambda(t \mid \boldsymbol{X}) = \lambda(t)\exp\{\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{X}(t) + \gamma X_j(t)I(t \leq y)\}.$$

We denote the resulting $W_n(x, y)$ and $\widehat{W}_n(x, y)$ as $U_{jn}(y)$ and $\widehat{U}_{jn}(y)$, respectively. One can conduct graphical inspection of the proportional hazards assumption by comparing the observed $U_{jn}(y)$ with a few simulated $\widehat{U}_{jn}(y)$ over the time index y. The p-value of the corresponding supremum test is obtained by generating a large number of realizations from $\sup_y |\widehat{U}_{jn}(y)|$ and comparing them to the observed value of $\sup_y |U_{jn}(y)|$.

We can check the overall fit of the model by choosing $g\{\boldsymbol{X}_i(t); \widehat{\boldsymbol{\beta}}, x, y\} = I\{\boldsymbol{X}_i(t) \leq x\}I(t \leq y)$. Then, model (3) is specified as

$$\lambda(t \mid \boldsymbol{X}) = \lambda(t)\exp[\boldsymbol{\beta}^{\mathrm{T}}\boldsymbol{X}(t) + \gamma I\{\boldsymbol{X}(t) \leq x\}I(t \leq y)].$$

Denote the corresponding $W_n(x, y)$ and $\widehat{W}_n(x, y)$ as $W_{n,o}(x, y)$ and $\widehat{W}_{n,o}(t, x)$, respectively. An omnibus test statistic is $\sup_{x,y} |W_{n,o}(x, y)|$, and a p-value can be obtained by comparing the observed $W_{n,o}(x, y)$ with the simulated $\widehat{W}_{n,o}(x, y)$ over the two indices x and y.

We state in Appendix B that the proposed supremum tests are consistent against the alternatives of interest. We display in Figure 1 some prototype mean functions for the model-checking processes under various forms of model misspecification. Panels (a), (b), (c), (d), (e), and (f) pertain to the situations in which the true effects of a covariate are quadratic, logarithmic, exponential, square-root, dichotomous, and cubic, respectively, and panels (g) and (h) pertain to the situations of monotonically decreasing and increasing hazard ratios, respectively. The resemblance of an observed pattern to a specific curve in Figure 1 would provide a useful hint on how to correct the misspecification. The residual patterns for the quadratic and exponential transformations are similar, and so are those of the logarithmic and square root transformations. We choose between two transformations with similar residual patterns using the Akaike information criterion.

3. Simulation Studies

We conducted simulation studies to evaluate the performance of the proposed methods. We simulated 10,000 replicated datasets with sample sizes of 200, 400, or 800 and generated six potential examination times for each subject by generating six time points from $\mathrm{Un}(0, 5)$ and letting U_{ij} be the jth ordered time point plus $\mathrm{Un}(0.1(j-1), 0.1j)$. We generated two independent covariates, a binary X_1 and a continuous X_2, and fit the model $\lambda(t)\exp(\beta_1 X_1 + \beta_2 X_2)$, with the convergence threshold in the EM algorithm set at 10^{-3}. Finally, we performed the supremum

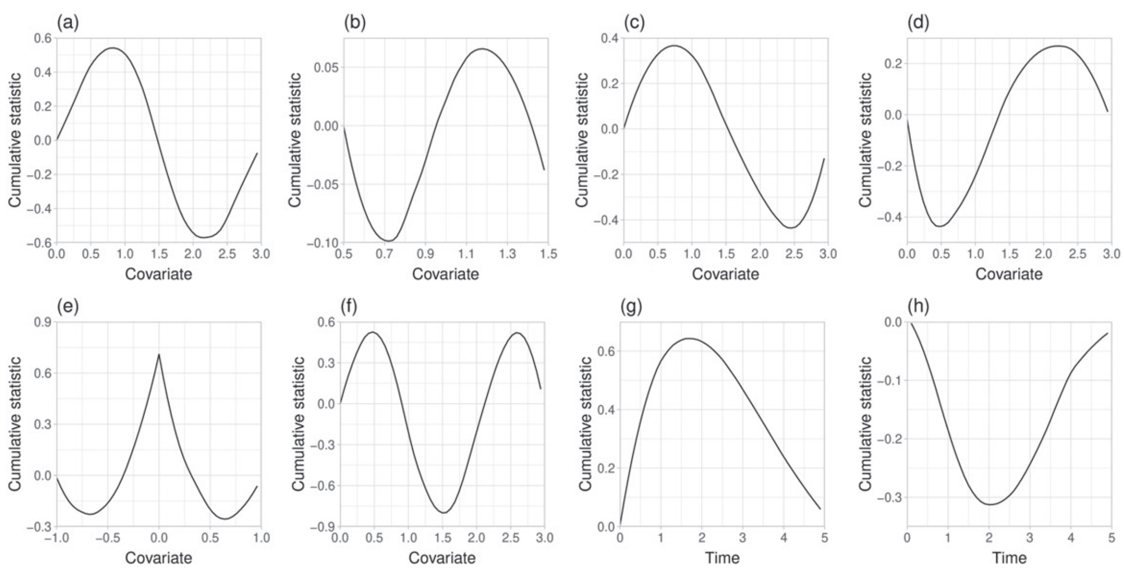

Figure 1. The mean functions of the model-checking processes when model (1) is misspecified: (a) $W_{1n}(x)$ when model $\lambda(t)\exp(\beta_1 X_1)$ is fitted to data generated by $\lambda(t)\exp(\beta_1 X_1 + \gamma X_1^2)$; (b) $W_{1n}(x)$ when model $\lambda(t)\exp(\beta_1 X_1)$ is fitted to data generated by $\lambda(t)\exp(\gamma \log X_1)$; (c) $W_{1n}(x)$ when model $\lambda(t)\exp(\beta_1 X_1)$ is fitted to data generated by $\lambda(t)\exp\{\gamma \exp(X_1)\}$; (d) $W_{1n}(x)$ when model $\lambda(t)\exp(\beta_1 X_1)$ is fitted to data generated by $\lambda(t)\exp(\gamma\sqrt{X_1})$; (e) $W_{1n}(x)$ when model $\lambda(t)\exp(\beta_1 X_1)$ is fitted to data generated by $\lambda(t)\exp\{\gamma I(X_1 < 0)\}$; (f) $W_{1n}(x)$ when model $\lambda(t)\exp(\beta_1 X_1 + \beta_2 X_1^2)$ is fitted to data generated by $\lambda(t)\exp(\beta_1 X_1 + \beta_2 X_1^2 + \gamma X_1^3)$; (g) $U_{1n}(y)$ when model $\lambda(t)\exp(\beta_1 X_1)$ is fitted to data generated by $\lambda(t)\exp(\gamma X_1 \log t)$; and (h) $U_{1n}(y)$ when model $\lambda(t)\exp(\beta_1 X_1)$ is fitted to data generated by $\lambda(t)\exp\{\gamma X_1 \log(1+t)\}$. The data in (a), (b), (c), (d), (e), (f), (g), and (h) are simulated with $\lambda(t) = t/8, t/8, t/8, t/50, 2t/9, 2t/9, t^{1/2}/5$ and $t^{1/2}/5$, $\gamma = 0.5, 1, 0.1, 3, 1, -1, -0.8$, and 0.8, and X_1 being $N(1.5, 1)$ truncated to $[0, 3]$, $\mathrm{Un}(0.5, 1.5)$, $\mathrm{Un}(0, 3)$, $\mathrm{Un}(0, 3)$, $\mathrm{Un}(-1, 1)$, $N(1.5, 1)$ truncated to $[0, 3]$, $\mathrm{Ber}(0.5)$, and $\mathrm{Ber}(0.5)$, respectively.

tests using 10,000 realizations of the Gaussian process with both δ_n and h_n set at $n^{-1/2}$.

In the first series of studies, we considered checking the functional form of a covariate. We let X_1 be Ber(0.5) and X_2 be $N(1.5, 1)$ truncated to $[0, 3]$. The true hazard function is $(t/8)\exp(0.5X_1 - 0.5X_2 + rX_2^2)$, where an additional quadratic term of X_2 is included, and r is a tuning parameter that controls how severely the functional form of X_2 is misspecified. When r increases from 0 to 0.8, the percentage of interval-censored observations increases from 55% to 70%, and the percentage of right-censored observations decreases from 40% to 5%. We checked the functional form of X_2 using the supremum test statistic $\sup_x |W_{2n}(x)|$. As a comparison, we performed a score test for testing no X_2^2 effect, the general procedure of which is described in Appendix C. The score test is optimal in this case because it directly tests the true alternative. The top panel of Figure 2 displays the empirical rejection rates of these tests under different combinations of n and r. The supremum test has proper control of the Type I error when $r = 0$. When $r > 0$, the power of the supremum test becomes close to that of the score test as the sample size increases. The graphical comparisons between $W_{2n}(x)$ and $\widehat{W}_{2n}(x)$ for one simulated dataset under $r = 0.8$ are provided in the top panel of Figure S1 in the supplementary materials, where the patterns of $W_{2n}(x)$ resemble the curves in panels (a) and (c) of Figure 1.

We next investigated a second type of misspecified functional form. We let X_1 be Ber(0.5) and X_2 be Un$(-1, 1)$ and specified the true hazard function as $(2t/9)\exp(0.5X_1 - 0.5X_2/|X_2|^r)$. In this way, the true functional form of X_2 can be viewed as a twisted version of X_2 with an S-shape. There are approximately 80% interval-censored observations and approximately 10% right-censored observations when r ranges from 0 to 1. We again compared the supremum test $\sup_x |W_{2n}(x)|$ with the score test for testing no X_2^2 effect, and the results are summarized in the bottom panel of Figure 2. The supremum test has proper control of the Type I error when $r = 0$ and reasonable power when $r > 0$, while the score test has essentially no power. The reason that the score test fails is that the true functional form is an odd function of X_2, such that adding a quadratic term will not provide additional information about the true model. The graphical comparisons between $W_{2n}(x)$ and $\widehat{W}_{2n}(x)$ for one simulated dataset under $r = 1$ are provided in the bottom panel of Figure S1 in the supplementary materials, where the patterns of $W_{2n}(x)$ resemble the curve in Figure 1(e).

We also evaluated the omnibus test $\sup_{x,y} |W_{n,o}(x, y)|$ in the above two scenarios, and the results are shown in Figure 2. When the model is correctly specified, the omnibus test properly controls the Type I error. In the first scenario, the power of the omnibus test is lower than that of the other two tests when the functional form of X_2 is misspecified. In the second scenario, the omnibus test is not as sensitive to misspecified functional form as the supremum test $\sup_x |W_{2n}(x)|$ is but still has much higher power than that of the score test.

In the second series of studies, we considered checking the proportional hazards assumption. We let X_1 be Ber(0.5) and X_2 be Un(0, 1) and specified the true hazard function as $0.2t^{1/2}\exp(rX_1 \log t + 0.5X_2)$. In this case, the hazard ratio with respect to X_1 is t^r, which is a constant when $r = 0$ and is a monotonically increasing function of t when $r > 0$. When r increases from 0 to 1, the percentage of interval-censored observations increases from 70% to 80%, and the percentage of right-censored observations decreases from 20% to 10%. We checked the proportional hazards assumption with respect to X_2 via the supremum test statistic $\sup_y |U_{1n}(y)|$ and performed the score test for testing no effects of $X_1 \log t$ or $X_1 t$ as comparisons. The score test for $X_1 \log t$ is optimal in this case because it directly tests the true alternative. As shown in the top panel of Figure 3, the supremum test has proper control of the Type

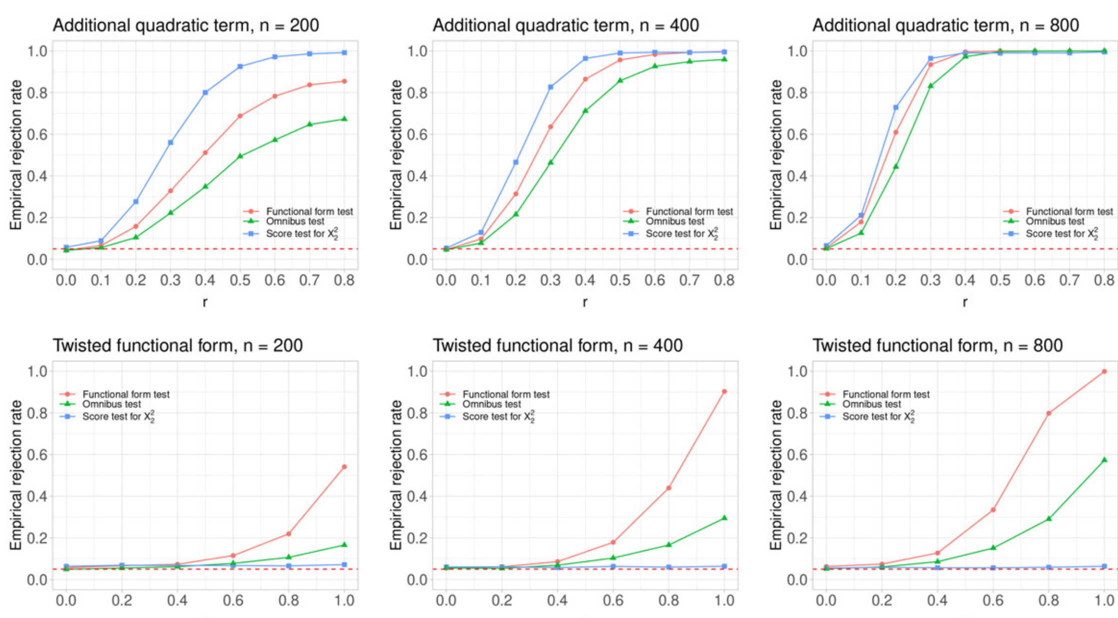

Figure 2. Empirical rejection rates of the supremum tests and the score test for X_2^2 when checking the functional form of X_2, as a function of the tuning parameter r. The curves with the circle, triangle, and square points pertain to $\sup_x |W_{2n}(x)|$, $\sup_{x,y} |W_{n,o}(x, y)|$ and the score test, respectively. The rejection rates are based on 10,000 realizations of the Gaussian processes. The dashed line represents the nominal level of the Type I error for all tests.

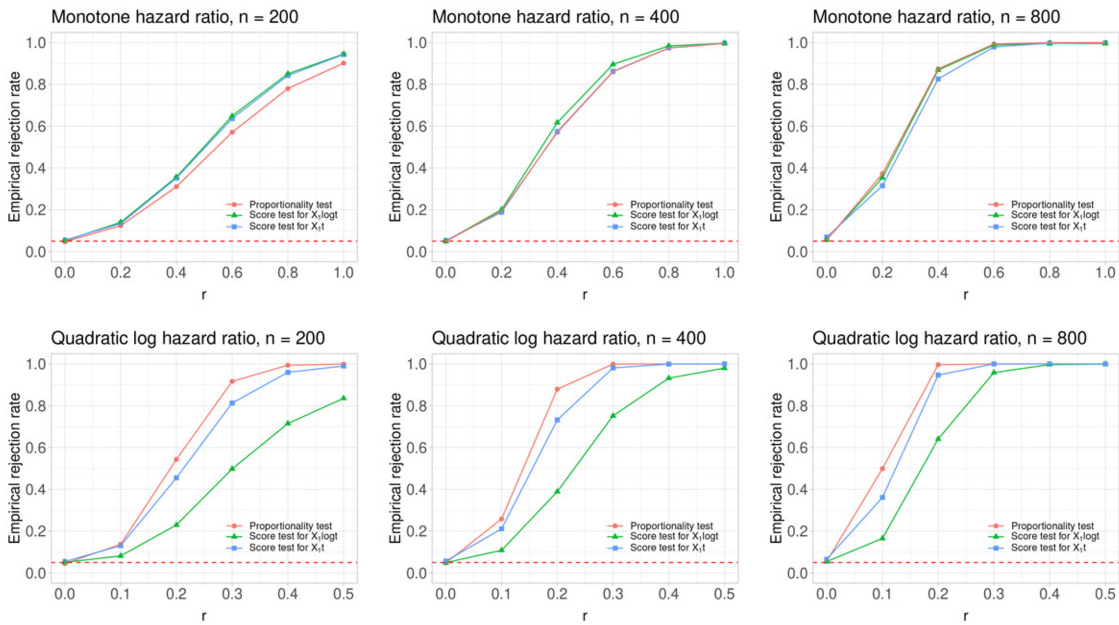

Figure 3. Empirical rejection rates of the supremum test $\sup_y |U_{1n}(y)|$ and the score tests for $X_1 \log t$ and $X_1 t$ when checking the proportional hazards assumption with respect to X_1, as a function of the tuning parameter r. The curves with the circle, triangle, and square points pertain to the supremum test, the score test for $X_1 \log t$ and the score test for $X_1 t$, respectively. The rejection rates are based on 10,000 realizations of the Gaussian processes. The dashed line represents the nominal level of the Type I error for all tests.

I error when $r = 0$ and similar power to that of the score tests when $r > 0$. The graphical comparisons between $U_{1n}(y)$ and $\widehat{U}_{1n}(y)$ for one simulated dataset under $r = 1$ are provided in the top panel of Figure S2 in the supplementary materials, where the patterns of $U_{1n}(y)$ resemble the curve in Figure 1(h).

We investigated a second type of nonproportional hazards by changing the true hazard function to $0.2t^{1/2} \exp[r\{(t-2)^2 - 3\}X_1 + 0.5X_2]$. In this way, the log hazard ratio with respect to X_1 is a quadratic function that first decreases, then increases. When r increases from 0 to 0.5, the percentage of interval-censored observations increases from 70% to 75%, and the percentage of right-censored observations decreases from 20% to 15%. We again compared the supremum test $\sup_y |U_{1n}(y)|$ with the score test, and the results are shown in the bottom panel of Figure 3. The supremum test always has higher power than that of the score tests when $r > 0$ and properly controls the Type I error when $r = 0$. The score tests do not perform well because they are designed to detect monotone hazard ratios. The graphical comparisons between $U_{1n}(y)$ and $\widehat{U}_{1n}(y)$ for one simulated dataset under $r = 1$ are provided in the bottom panel of Figure S2 in the supplementary materials, where the patterns of $U_{1n}(y)$ resemble the curve in Figure 1(h).

4. Application

We considered the Atherosclerosis Risk in Communities Study, which involves a cohort of 15,792 Caucasian and African-American individuals from four U.S. communities: Forsyth County, North Carolina; Jackson, Mississippi; suburbs of Minneapolis, Minnesota; and Washington County, Maryland (Wright et al. 2021). The participants received a baseline examination in 1987–1989, three follow-up examinations at approximately three-year intervals, and a further examination in 2011–2013. A major objective of the study was to investigate risk factors for diabetes and hypertension; both conditions were monitored at the examination times and thus interval-censored. We related the incidence of diabetes and hypertension to race, gender, communities and five baseline risk factors: age, body mass index, glucose level, systolic blood pressure and diastolic blood pressure. All the covariates were measured at baseline. The analysis dataset consists of a total of 8735 individuals without prior diabetes or hypertension with 2235 or 2293 distinct interval endpoints for diabetes or hypertension, respectively. To avoid the impact of model selection on statistical inference, we equally divided the dataset into two parts: one for model selection and the other for inference.

With the first half of the data, we fitted model (1) to the endpoint of diabetes with the aforementioned covariates. We checked the proportional hazards assumption with respect to each covariate via the proposed graphical inspection and supremum test. The left panel of Figure 4 displays the process $U_{jn}(y)$ versus follow-up time for baseline glucose level along with 20 simulated processes $\widehat{U}_{jn}(y)$. The observed process deviates substantially from its null distribution, and the p-value of the supremum test is smaller than 0.001, revealing violation of the proportional hazards assumption. The same procedure did not detect nonproportional hazards for other covariates. The observed pattern for baseline glucose level resembles the curve in Figure 1(g), suggesting a monotonically decreasing hazard ratio. After adding the product of baseline glucose level and the logarithm of follow-up time, we checked the proportional hazards assumption with respect to baseline glucose level again and obtained a p-value close to 1.

Given the revised model, we further checked the functional forms of all continuous covariates: age, body mass index, glucose level, systolic blood pressure and diastolic blood pressure. The right panel of Figure 4 plots the process $W_{jn}(x)$ against baseline body mass index along with 20 simulated processes $\widehat{W}_{jn}(x)$. The

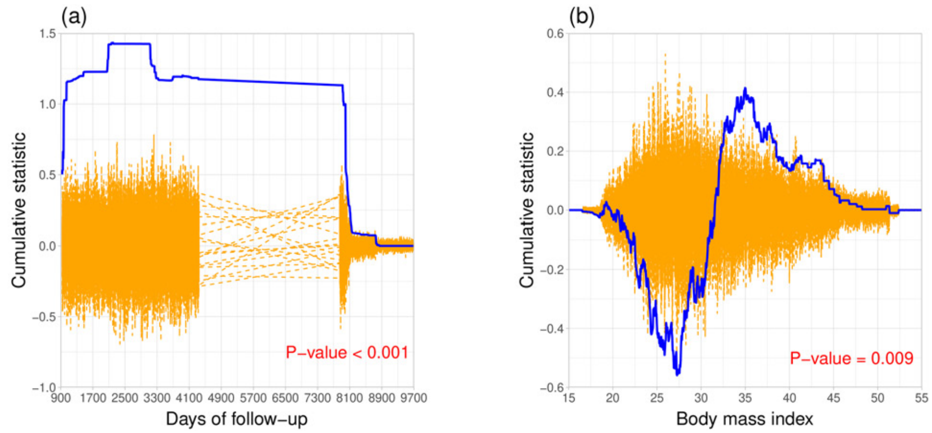

Figure 4. Plots of the model-checking processes for the endpoint of diabetes: (a) $U_{jn}(y)$ and $\widehat{U}_{jn}(y)$ against follow-up time for baseline glucose level; (b) $W_{jn}(x)$ and $\widehat{W}_{jn}(x)$ against baseline body mass index. The solid and dashed curves pertain to the observed processes and the simulated processes, respectively.

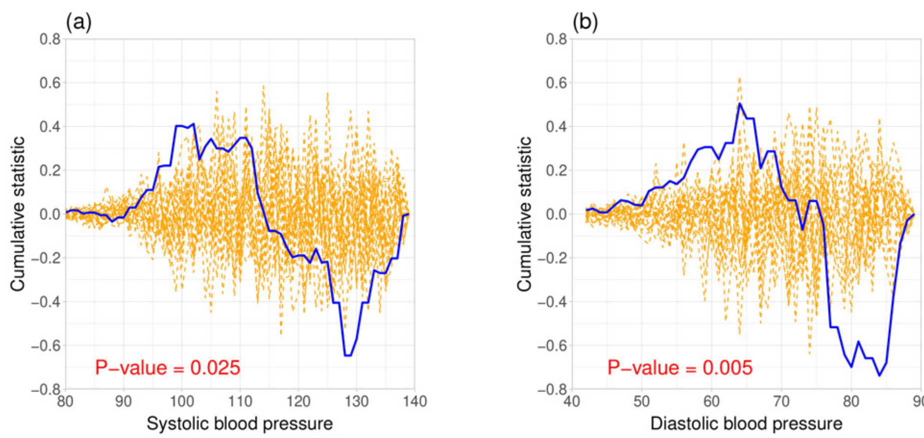

Figure 5. Plots of the model-checking processes for the endpoint of hypertension: (a) $W_{jn}(x)$ and $\widehat{W}_{jn}(x)$ against baseline systolic blood pressure; (b) $W_{jn}(x)$ and $\widehat{W}_{jn}(x)$ against baseline diastolic blood pressure. The solid and dashed curves pertain to the observed processes and the simulated processes, respectively.

observed process deviates from its null distribution with the corresponding p-value of 0.009, indicating that the functional form of baseline body mass index might be misspecified. The same procedure did not detect misspecification for other covariates. The observed pattern for baseline body mass index resembles the curves in panels (b) and (d) of Figure 1. The Akaike information criterion selected the logarithmic transformation over the square root transformation. Therefore, we further revised the model by replacing body mass index with log(body mass index) and checked its functional form again. The p-value increased to 0.286, implying that the logarithmic transformation is a better functional form for this covariate. For this model, we checked its exponential link function via $\sup_x |W_{n,l}(x)|$. The supremum test generated a p-value of 0.972, indicating no significant evidence of misspecification.

Finally, we fit this final model to the other half of the data. The parameter estimates for baseline glucose value and its corresponding time-dependent covariate are 0.512 and −0.053, respectively, with the respective estimated standard errors being 0.010 and 0.001. Therefore, the time-varying effect of baseline glucose value is very significant, consistent with what we found in the model-checking process. The parameter estimate for log(body mass index) is 1.916, with an estimated standard error of 0.237. Under this model, the hazard ratio of baseline glucose value decreases from 1.307 at 100 days to 1.030 at 9000 days; the increment in the log hazard ratio of body mass index is 0.093 for a one-unit increase in body mass index at 20 kg/m^2 but becomes 0.038 for a one-unit increase in body mass index at 50 kg/m^2. In contrast, the model with the proportional hazards assumption of baseline glucose value would estimate the hazard ratio at 1.102 over time, and the model with a linear form of body mass index would estimate the increment in the log hazard ratio at 0.060 for a one-unit increase in body mass index at any level.

We observed a different type of misspecified functional form when analyzing the other endpoint, hypertension. We fitted model (1) with all baseline covariates in their original forms and checked the functional form of each continuous covariate. Figure 5 displays the process $W_{jn}(x)$ for baseline systolic or diastolic blood pressure along with 20 simulated processes $\widehat{W}_{jn}(x)$. The observed processes deviate substantially from their null distributions with very small p-values, indicating misspecified functional forms of the two covariates. The observed patterns resemble the curves in panels (a) and (c) of Figure 1. The Akaike information criterion selected the quadratic transformation over the exponential transformation. After adding (systolic blood pressure)^2 and (diastolic blood pressure)^2, the p-values of the supremum tests increased to 0.959 and 0.226 for systolic and diastolic blood pressures, respectively. We checked other aspects of this model and improved its goodness of fit in the same way as we did for diabetes.

5. Discussion

Martingale residuals (e.g., Barlow and Prentice 1988; Therneau, Grambsch, and Fleming 1990; Lin, Wei, and Ying 1993), which have been widely used to check the Cox proportional hazards model with right-censored data, are not well-defined under interval censoring. To circumvent this difficulty, we developed an alternative approach by expressing the stochastic processes for checking the model assumptions in terms of the score statistics for testing zero regression parameters in an expanded family of Cox models. The proposed statistics reduce to the cumulative sums of martingale residuals (Lin, Wei, and Ying 1993) in the special case of right-censored data. Due to the lack of an analytic expression for the profile likelihood under interval censoring, numerical derivatives and the EM algorithm were used to compute the score statistics, and proving the weak convergence of the stochastic processes and the consistency of the supremum tests required new mathematical arguments. In addition, we went further than the existing literature on right-censored data in providing guidance on how to correct model misspecification.

The proposed approach can be extended to other semiparametric regression models under interval censoring (Zeng, Gao, and Lin 2017; Zeng and Lin 2020; Xu, Zeng, and Lin 2022, 2024). Specifically, Zeng, Gao, and Lin (2017) and Zeng and Lin (2020) studied random-effects models for multivariate interval-censored data and panel count data, respectively. The test statistics for checking those models can be constructed by taking the numerical difference of the profile log-likelihood function, where the random effects are integrated out. The extensions to the marginal models of Xu, Zeng, and Lin (2022, 2024) are more straightforward.

The proposed model-checking procedures accommodate time-dependent covariates and require the covariate values at the unique interval endpoints. While time-dependent covariates may not be measured at all unique interval endpoints in practice, extrapolation can be used to determine the unmeasured values, making the computation feasible. To formally account for intermittent missing values or measurement errors, we can extend the proposed approach by jointly modeling the covariate process and the event time (Zeng and Lin 2007). In the joint model, the observed covariates are treated as repeated measures arising from a latent process with potential measurement error, and the latent process is incorporated into the event time model as a time-dependent covariate. The model-checking procedures can be developed based on the score test statistics of the joint model.

The assumption that the examination times are independent of the event time conditional on the covariates can be assessed using the score test described in Appendix C. Specifically, let $Z(t) = \sum_{l=1}^{K} I(U_l \leq t)$ denote the counting process of the examination times, and the score statistic for testing no effect of $Z(t)$ on T can be used to assess the independent censoring assumption. We applied this score test to the diabetes and hypertension endpoints in Section 4. The resulting p-values are 0.653 and 0.843 for diabetes and hypertension, respectively, providing no evidence against the independent censoring assumption. Alternatively, the independent censoring assumption can be assessed by jointly modeling the examination times and the event time, where the potential dependence is characterized by a random effect. The score statistic of the joint model for testing zero variance of the random effect provides a formal test for the independent censoring assumption.

The proposed omnibus test requires evaluating $W_n(x, y)$ and $\widehat{W}_n(x, y)$ at all unique values of the indices x and y, which can be time-consuming. To reduce computational burden, we suggest using only a subset of the unique values. The subset can be chosen as a sufficiently dense grid over the ranges of x and y. The computation can be paralleled because the evaluation of $W_n(x, y)$ and $\widehat{W}_n(x, y)$ at one index does not affect their evaluation at a different index.

As a by-product, we derived the score statistic for testing zero regression parameters under model (1) with interval-censored data. It can be viewed as an extension of the two-sample nonparametric test statistic of Zhang, Liu, and Zhan (2001). The score test statistic can be similarly derived for the proportional rates model with panel count data (Xu, Zeng, and Lin 2024) and serves as an extension of the nonparametric k-sample test statistic of Zhang (2006).

Appendix A: Equivalence of the Test Statistics and the Score Statistics

If there is no censoring, the full log-likelihood function under model (3) is

$$l(\gamma, \boldsymbol{\beta}, \Lambda) = \sum_{i=1}^{n} \Bigg(\log \lambda(T_i) + \boldsymbol{\beta}^{\mathrm{T}} \boldsymbol{X}_i(T_i) + \gamma g\{\boldsymbol{X}_i(T_i); \boldsymbol{\beta}, x, y\} - \int_0^{T_i} \exp[\boldsymbol{\beta}^{\mathrm{T}} \boldsymbol{X}_i(u) + \gamma g\{\boldsymbol{X}_i(u); \boldsymbol{\beta}, x, y\}] \mathrm{d}\Lambda(u) \Bigg).$$

Under the null hypothesis of $\gamma = 0$, $l(0, \boldsymbol{\beta}, \Lambda)$ is maximized at $(\widehat{\boldsymbol{\beta}}_p, \widehat{\Lambda}_p)$, where $\widehat{\boldsymbol{\beta}}_p$ is the partial maximum likelihood estimator (Cox 1975) and

$$\widehat{\Lambda}_p(t) = \int_0^t \frac{\sum_{i=1}^n \mathrm{d}N_i(u)}{\sum_{i=1}^n I(T_i \geq u) \exp\{\widehat{\boldsymbol{\beta}}_p^{\mathrm{T}} \boldsymbol{X}_i(u)\}}$$

is the Breslow (1972) estimator. Taking the derivative of $l(\gamma, \boldsymbol{\beta}, \Lambda)$ with respect to γ, the score function for γ under model (3) is given by

$$U(\gamma; \boldsymbol{\beta}, \Lambda) = \sum_{i=1}^{n} \Bigg(g\{\boldsymbol{X}_i(T_i); \boldsymbol{\beta}, x, y\} - \int_0^{T_i} g\{\boldsymbol{X}_i(u); \boldsymbol{\beta}, x, y\} \exp[\boldsymbol{\beta}^{\mathrm{T}} \boldsymbol{X}_i(u) + \gamma g\{\boldsymbol{X}_i(u); \boldsymbol{\beta}, x, y\}] \mathrm{d}\Lambda(u) \Bigg).$$

Setting $(\gamma, \boldsymbol{\beta}, \Lambda)$ to $(0; \widehat{\boldsymbol{\beta}}_p, \widehat{\Lambda}_p)$, the score statistics for testing the null hypothesis of $\gamma = 0$ under model (3) is

$$U(0; \widehat{\boldsymbol{\beta}}_p, \widehat{\Lambda}_p) = \sum_{i=1}^{n} \int_0^{\infty} \Bigg[g\{\boldsymbol{X}_i(u); \widehat{\boldsymbol{\beta}}_p, x, y\} - \frac{\sum_{j=1}^n I(T_j \geq u) \exp\{\widehat{\boldsymbol{\beta}}_p^{\mathrm{T}} \boldsymbol{X}_j(u)\} g\{\boldsymbol{X}_j(u); \widehat{\boldsymbol{\beta}}_p, x, y\}}{\sum_{j=1}^n I(T_j \geq u) \exp\{\widehat{\boldsymbol{\beta}}_p^{\mathrm{T}} \boldsymbol{X}_j(u)\}} \Bigg] \mathrm{d}N_i(u).$$

By rearranging the summation symbols, $U(0;\widehat{\boldsymbol\beta}_p,\widehat{\Lambda}_p)$ simplifies to

$$\sum_{i=1}^{n}\int_0^\infty g\{\boldsymbol X_i(u);\widehat{\boldsymbol\beta}_p,x,y\}\mathrm d\widehat{M}_i(u),$$

where $\widehat{M}_i(t)=N_i(t)-\int_0^t I(T_i\ge u)\exp\{\widehat{\boldsymbol\beta}_p^{\mathrm T}\boldsymbol X_i(u)\}\mathrm d\widehat{\Lambda}_p(u)$ $(i=1,\dots,n)$ are the martingale residuals. Thus, $n^{-1}U(0;\widehat{\boldsymbol\beta}_p,\widehat{\Lambda}_p)$ is the empirical average of the statistic in (2) with $\boldsymbol\beta$ and Λ replaced by their maximum likelihood estimators.

Appendix B: Consistency of Supremum Tests

We assume that the Kullback-Leibler discrepancy $E[\log\{f(0,\Lambda;\boldsymbol\beta,x,y)/f_0\}]$ has a unique maximizer, denoted by $(\boldsymbol\beta^*,\Lambda^*)$. Here, $f(\gamma,\Lambda;\boldsymbol\beta,x,y)$ is the likelihood function of a single observation O under model (3), and f_0 is the true likelihood of O. We state below the results on the consistency of the supremum tests.

1. *Omnibus test.* The supremum test $\sup_{x,y}|W_{n,o}(x,y)|$ is consistent for testing the null hypothesis that $\lambda(t\mid\boldsymbol X)=\lambda(t)\exp\{\boldsymbol\beta^{\mathrm T}\boldsymbol X(t)\}$ for some $\boldsymbol\beta$ and $\lambda(\cdot)$ against the general alternative hypothesis that there do not exist a vector $\boldsymbol\beta_0$ and a function $\lambda_0(\cdot)$ such that $\lambda(t\mid\boldsymbol X)=\lambda_0(t)\exp\{\boldsymbol\beta_0^{\mathrm T}\boldsymbol X(t)\}$ for all $t\in[0,\tau]$.
2. *Functional forms of covariates.* Suppose that $\boldsymbol X$ is time-independent. The supremum test $\sup_x|W_{jn}(x)|$ is consistent against the alternative hypothesis that $\lambda(t\mid\boldsymbol X)=\lambda^*(t)\exp(\boldsymbol\beta_{-j}^{*\mathrm T}\boldsymbol X_{-j})\mu_0(X_j)$ for some function $\mu_0(\cdot)>0$ but there does not exist a constant α such that $\mu_0(x)/\exp(\alpha x)$ is constant. Here, X_j is the jth component of $\boldsymbol X$, $\boldsymbol X_{-j}$ consists of the other components of $\boldsymbol X$, $\boldsymbol\beta_{-j}^*$ consists of the components of $\boldsymbol\beta^*$ corresponding to $\boldsymbol X_{-j}$, and $\lambda^*(t)$ is the derivative of $\Lambda^*(t)$.
3. *Link function.* Suppose that $\boldsymbol X$ is time-independent. The supremum test $\sup_x|W_{n,l}(x)|$ is consistent against the alternative hypothesis that $\lambda(t\mid\boldsymbol X)=\lambda^*(t)L_0(\boldsymbol\beta^{*\mathrm T}\boldsymbol X)$ for some function $L_0(\cdot)>0$, where $L_0(\cdot)$ is not exponential.
4. *Proportionality.* Suppose that there exists a deterministic function $x_j(t)\neq 0$ such that $\Pr\{X_j(t)=x_j(t)\}=\xi>0$ and $\Pr\{X_j(t)=0\}=1-\xi$. The supremum test $\sup_y|U_{jn}(y)|$ is consistent against the alternative hypothesis that $\lambda(t\mid\boldsymbol X)=\lambda^*(t)\exp\{\boldsymbol\beta_{-j}^{*\mathrm T}\boldsymbol X_{-j}(t)+\beta_{j0}(t)X_j(t)\}$ for some nonconstant function $\beta_{j0}(\cdot)$.

We assume that the Kullback-Leibler discrepancy $E[\log\{f(\gamma,\Lambda;\boldsymbol\beta^*,x,y)/f_0\}]$ has a unique maximum at (γ_e^*,Λ_e^*) for each fixed $(x,y)\in\mathbb T$. In addition, we assume that when $\gamma_e^*\neq 0$ for some $(x_0,y_0)\in\mathbb T$, $W_n(x_0,y_0)$ converges weakly to a noncentral distribution. Therefore, establishing the consistency of the supremum tests is equivalent to proving that γ_e^* is nonzero for some (x_0,y_0) under the alternatives. If $\gamma_e^*=0$ for any $(x,y)\in\mathbb T$, then $(0,\Lambda^*)$ is the unique maximizer of $\mathbb Pl(\gamma,\Lambda;\boldsymbol\beta^*,x,y)$ for any (x,y), which implies that $\mathbb Pl_\gamma(0,\Lambda^*;\boldsymbol\beta^*,x,y)=0$ for any (x,y). Denote the true survival function of T by $H_0(t)=\exp[-\int_0^t h_0\{u;\boldsymbol X(u)\}\mathrm du]$. Write $H_1(t)=H(t;\boldsymbol\beta^*,0,\Lambda^*,x,y)$. Then $\mathbb Pl_\gamma(0,\Lambda^*;\boldsymbol\beta^*,x,y)$ can be written as

$$\begin{aligned}
&E\Bigg[\sum_{l=0}^{K}\sum_{m=0}^{K}\int_0^\tau g\{\boldsymbol X(u);\boldsymbol\beta^*,x,y\}\exp\{\boldsymbol\beta^{*\mathrm T}\boldsymbol X(u)\}I(U_m<u\le U_{m+1})\\
&\quad\times\frac{H_0(U_l)-H_0(U_{l+1})}{H_1(U_l)-H_1(U_{l+1})}\{H_1(U_{l+1})I(u\le U_{l+1})\\
&\quad-H_1(U_l)I(u\le U_l)\}\mathrm d\Lambda^*(u)\Bigg]\\
&=-E\Bigg[\sum_{l=0}^{K}\sum_{m=0}^{l-1}\{H_0(U_l)-H_0(U_{l+1})\}\int_{U_m}^{U_{m+1}}g\{\boldsymbol X(u);\boldsymbol\beta^*,x,y\}\\
&\quad\times\exp\{\boldsymbol\beta^{*\mathrm T}\boldsymbol X(u)\}\mathrm d\Lambda^*(u)\Bigg]\\
&\quad+E\Bigg[\sum_{l=0}^{K}\frac{H_0(U_l)-H_0(U_{l+1})}{H_1(U_l)-H_1(U_{l+1})}H_1(U_{l+1})\int_{U_l}^{U_{l+1}}g\{\boldsymbol X(u);\boldsymbol\beta^*,x,y\}\\
&\quad\times\exp\{\boldsymbol\beta^{*\mathrm T}\boldsymbol X(u)\}\mathrm d\Lambda^*(u)\Bigg]\\
&=E\Bigg[\sum_{l=0}^{K-1}\frac{H_0(U_l)H_0(U_{l+1})}{H_1(U_l)-H_1(U_{l+1})}\left\{\frac{H_1(U_{l+1})}{H_0(U_{l+1})}-\frac{H_1(U_l)}{H_0(U_l)}\right\}\\
&\quad\times\int_{U_l}^{U_{l+1}}g\{\boldsymbol X(u);\boldsymbol\beta^*,x,y\}\exp\{\boldsymbol\beta^{*\mathrm T}\boldsymbol X(u)\}\mathrm d\Lambda^*(u)\Bigg].
\end{aligned}$$

By the mean-value theorem, there exists a random variable V_l such that

$$\frac{H_1(U_{l+1})}{H_0(U_{l+1})}-\frac{H_1(U_l)}{H_0(U_l)}=-\exp(V_l)\int_{U_l}^{U_{l+1}}D\{u;\boldsymbol X(u)\}\mathrm du$$

for $l=0,\dots,K-1$, where $D\{u;\boldsymbol X(u)\}=\lambda^*(u)\exp\{\boldsymbol\beta^{*\mathrm T}\boldsymbol X(u)\}-h_0\{u;\boldsymbol X(u)\}$. It follows that

$$E\Bigg[\sum_{l=0}^{K-1}W_l\int_{U_l}^{U_{l+1}}D\{u;\boldsymbol X(u)\}\mathrm du\int_{U_l}^{U_{l+1}}g\{\boldsymbol X(u);\boldsymbol\beta^*,x,y\}\lambda^*(u)\exp\{\boldsymbol\beta^{*\mathrm T}\boldsymbol X(u)\}\mathrm du\Bigg]=0$$

for any (x,y), where $W_l=-\exp(V_l)\frac{H_0(U_l)H_0(U_{l+1})}{H_1(U_l)-H_1(U_{l+1})}<0$ with probability 1. We only need to show that $D\{t;\boldsymbol X(t)\}$ is in the L_2-closure of the function class $\{g\{\boldsymbol X(t);\boldsymbol\beta^*,x,y\}\lambda^*(t)\times\exp\{\boldsymbol\beta^{*\mathrm T}\boldsymbol X(t)\}\}$ because in this case we can obtain that $\int_{U_l}^{U_{l+1}}D\{u;\boldsymbol X(u)\}\mathrm du=0$ with probability 1 for $l=0,\dots,K-1$. Then, $h_0\{t;\boldsymbol X(t)\}=\lambda^*(t)\exp\{\boldsymbol\beta^{*\mathrm T}\boldsymbol X(t)\}$ for all t with probability 1, which contradicts the alternative.

We first consider the omnibus test. For any bivariate function $h_0\{t;\boldsymbol X(t)\}$, there exists a sequence of constants (a_m,x_{1m},y_{1m}) such that $h_0\{t;\boldsymbol X(t)\}\lambda^{*-1}(t)\exp\{-\boldsymbol\beta^{*\mathrm T}\boldsymbol X(t)\}=\lim_{N\to\infty}\sum_{m=1}^N a_mI\{\boldsymbol X(t)\le x_{1m}\}I(t\le y_{1m})$ with probability 1. Therefore, $D\{t;\boldsymbol X(t)\}$ is in the L_2-closure of the function class $\{I\{\boldsymbol X(t)\le x\}I(t\le y)\lambda^*(t)\exp\{\boldsymbol\beta^{*\mathrm T}\boldsymbol X(t)\}\}$. For checking the functional form of X_j, it suffices to note that there exists a sequence of constants (b_m,x_{2m}) such that $\mu_0(X_j)\exp(-\beta_j^*X_j)=\lim_{N\to\infty}\sum_{m=1}^N b_mI(X_j\le x_{2m})$ with probability 1, where β_j^* is the jth component of $\boldsymbol\beta^*$. The same arguments can be applied to the test for checking the exponential link function. For checking the proportional hazards assumption,

$$\begin{aligned}
D\{t;\boldsymbol X(t)\}\,|_{X_j=x_j}&=\lambda^*(t)x_j(t)\{\beta_j^*-\beta_{j0}(t)\}\exp\{\boldsymbol\beta_{-j}^{*\mathrm T}\boldsymbol X_{-j}(t)\\
&\quad+\widetilde{\beta}_{j0}(t)x_j(t)\}
\end{aligned}$$

by the mean-value theorem, where $\widetilde{\beta}_{j0}(t)$ is a convex combination of β_j^* and $\beta_{j0}(t)$. There exists a sequence of constants (c_m,y_{2m}) such that $\{\beta_j^*-\beta_{j0}(t)\}\exp[\{\widetilde{\beta}_{0j}(t)-\beta_j^*\}x_j(t)]=\lim_{N\to\infty}\sum_{m=1}^N c_mI(t\le y_{2m})$.

Appendix C: Score Test Statistics

We describe below the score statistics for testing zero regression parameters. We consider the following expanded Cox model

$$\lambda(t\mid\boldsymbol X)=\lambda(t)\exp\{\boldsymbol\beta^{\mathrm T}\boldsymbol X(t)+\boldsymbol\gamma^{\mathrm T}\boldsymbol Z(t)\},\qquad\text{(C1)}$$

where $\boldsymbol{Z}(t) \in \mathbb{R}^q$ is q-vector of covariates not included in the fitted model. Suppose that we are interested in $H_0 : \boldsymbol{\gamma} = \boldsymbol{0}$. We consider the expanded likelihood function

$$L_n^*(\boldsymbol{\gamma}, \Lambda; \widehat{\boldsymbol{\beta}}) = \prod_{i=1}^n \left(\exp\left[- \int_0^{L_i} \exp\{\widehat{\boldsymbol{\beta}}^{\mathrm{T}} \boldsymbol{X}_i(u) + \boldsymbol{\gamma}^{\mathrm{T}} \boldsymbol{Z}_i(t)\} \mathrm{d}\Lambda(u) \right] - \exp\left[- \int_0^{R_i} \exp\{\widehat{\boldsymbol{\beta}}^{\mathrm{T}} \boldsymbol{X}_i(u) + \boldsymbol{\gamma}^{\mathrm{T}} \boldsymbol{Z}_i(t)\} \mathrm{d}\Lambda(u) \right] \right)$$

and calculate the profile loglikelihood function $\mathrm{pl}_n^*(\boldsymbol{\gamma}; \widehat{\boldsymbol{\beta}}) = \max_\Lambda \log L_n^*(\boldsymbol{\gamma}, \Lambda; \widehat{\boldsymbol{\beta}})$ using the EM-type algorithm.

The score test is based on the derivative of $\mathrm{pl}_n^*(\boldsymbol{\gamma}; \widehat{\boldsymbol{\beta}})$ with respect to $\boldsymbol{\gamma}$ at $\boldsymbol{\gamma} = \boldsymbol{0}$, which can be calculated by $\boldsymbol{W}_n^* = n^{-1/2} D_{h_n} \mathrm{pl}_n^*(\boldsymbol{0}; \widehat{\boldsymbol{\beta}})$, where D_{h_n} is the first-order numerical derivative for $\boldsymbol{\gamma} \in \mathbb{R}^q$ with perturbation constant h_n of order $n^{-1/2}$. We can analogously define the efficient Fisher information $\boldsymbol{V}_0^*$ for model (C1) and show that $\boldsymbol{W}_n^*$ is asymptotically normal with a limiting variance of $\boldsymbol{V}^*_{\boldsymbol{\gamma}_0\boldsymbol{\gamma}_0} - \boldsymbol{V}^{*\mathrm{T}}_{\boldsymbol{\beta}_0\boldsymbol{\gamma}_0} \boldsymbol{V}^{*-1}_{\boldsymbol{\beta}_0\boldsymbol{\beta}_0} \boldsymbol{V}^*_{\boldsymbol{\beta}_0\boldsymbol{\gamma}_0}$. Thus, under H_0, $\boldsymbol{W}_n^{*\mathrm{T}} (\boldsymbol{V}^*_{\boldsymbol{\gamma}_0\boldsymbol{\gamma}_0} - \boldsymbol{V}^{*\mathrm{T}}_{\boldsymbol{\beta}_0\boldsymbol{\gamma}_0} \boldsymbol{V}^{*-1}_{\boldsymbol{\beta}_0\boldsymbol{\beta}_0} \boldsymbol{V}^*_{\boldsymbol{\beta}_0\boldsymbol{\gamma}_0})^{-1} \boldsymbol{W}_n^*$ converges in distribution to a Chi-squared distribution with q degrees of freedom. Then, the score test statistic is $S_n = \boldsymbol{W}_n^{*\mathrm{T}} (\widehat{\boldsymbol{V}}^*_{\boldsymbol{\gamma}_0\boldsymbol{\gamma}_0} - \widehat{\boldsymbol{V}}^{*\mathrm{T}}_{\boldsymbol{\beta}_0\boldsymbol{\gamma}_0} \widehat{\boldsymbol{V}}^{*-1}_{\boldsymbol{\beta}_0\boldsymbol{\beta}_0} \widehat{\boldsymbol{V}}^*_{\boldsymbol{\beta}_0\boldsymbol{\gamma}_0})^{-1} \boldsymbol{W}_n^*$, where the limiting variance is replaced by its empirical counterpart.

Supplementary Materials

The supplementary materials contain the proofs of Theorems 1–3 and additional simulation results.

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